Code-Layout, Readability and Reusability

Date	08 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:
This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used consistent 4-space indentation throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into separate folders for source code, utilities, assets, reports, and models.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables use descriptive names that clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named according to their functionality using standard naming conventions.
5	Code Comments	Inline and block comments explain log	Yes	Important functions and complex logic include comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	Speech processing, semantic evaluation, UI,

				PDF generation, and utilities are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed by creating reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles invalid inputs, missing audio files, and runtime exceptions gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Speech-to-Text Module	Python (Whisper)	Audio transcription	High
2	Semantic Similarity Module	Python (Sentence-BERT)	Answer evaluation	High
3	Audio Feature Extraction Module	Python (Librosa)	Fluency and clarity analysis	High
4	PDF Report Generator	Python (ReportLab/FPDF)	Evaluation report generation	Medium
5	Streamlit UI Components	Streamlit	User interface pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a logical folder hierarchy.
Readability	5	Clear naming conventions, proper formatting, and comments improve readability.
Reusability	5	Core functionalities are implemented as reusable modules and functions.
Documentation / Comments	4	Key functions are documented with sufficient comments.
Overall Score	4.75	The project demonstrates good coding standards, modularity, and maintainability suitable for future enhancements.